

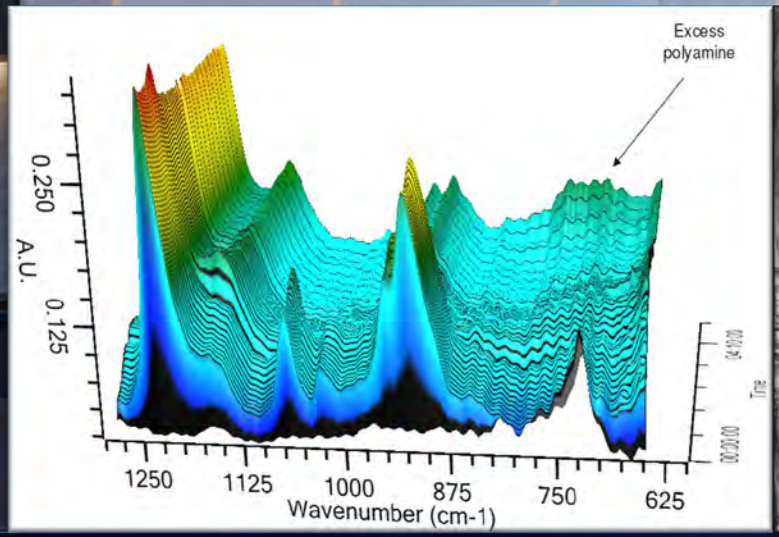




# Process-Control in-line with Spectral Fiber Systems

*Slava Artyushenko, Boris Mizaikoff, Martin Warman, Brad Swarbrick, Polina Fomina, Alex Novikov*








# Spectroscopy Market forecast

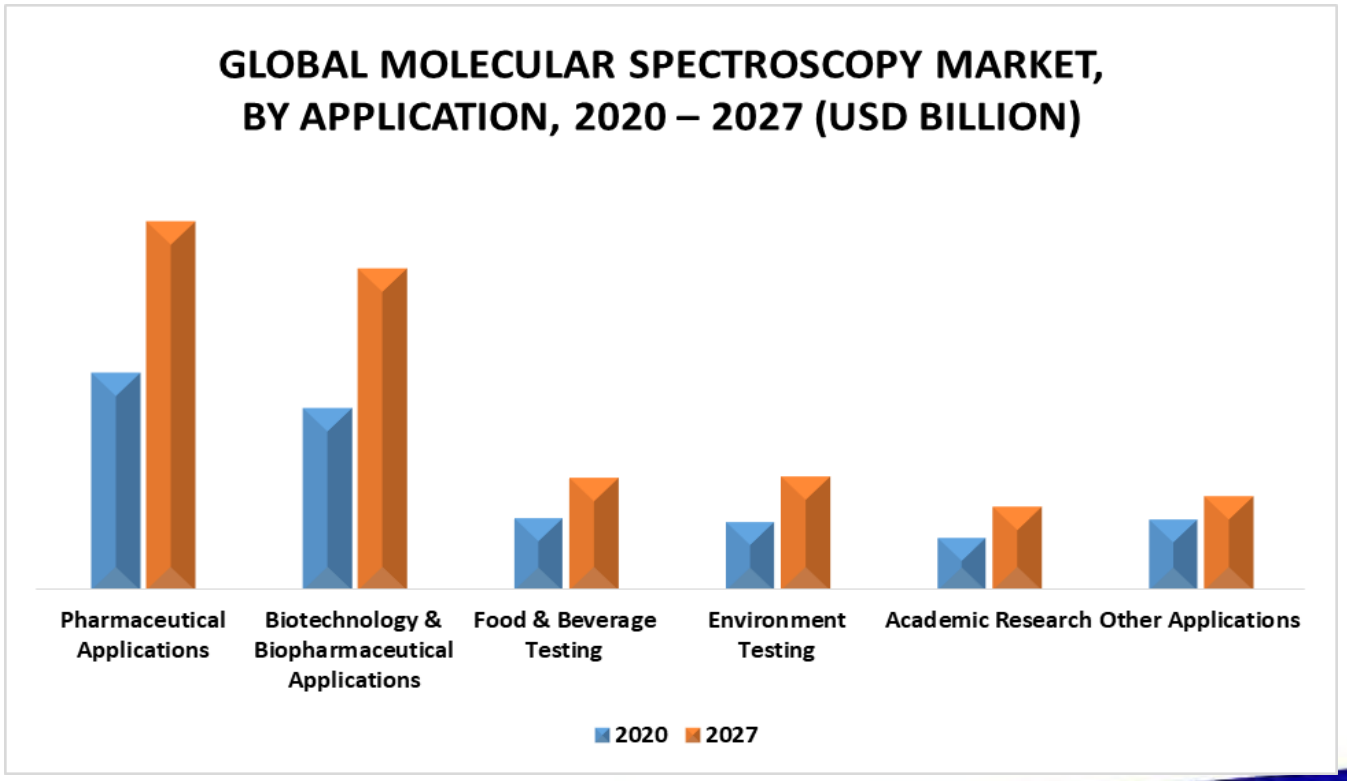
art photonics

## Main drivers of the growth this market:

- food safety concerns
- growth of pharmaceutical & biotechnology industry
- technological advancements in molecular spectroscopy & their application in environmental screening.





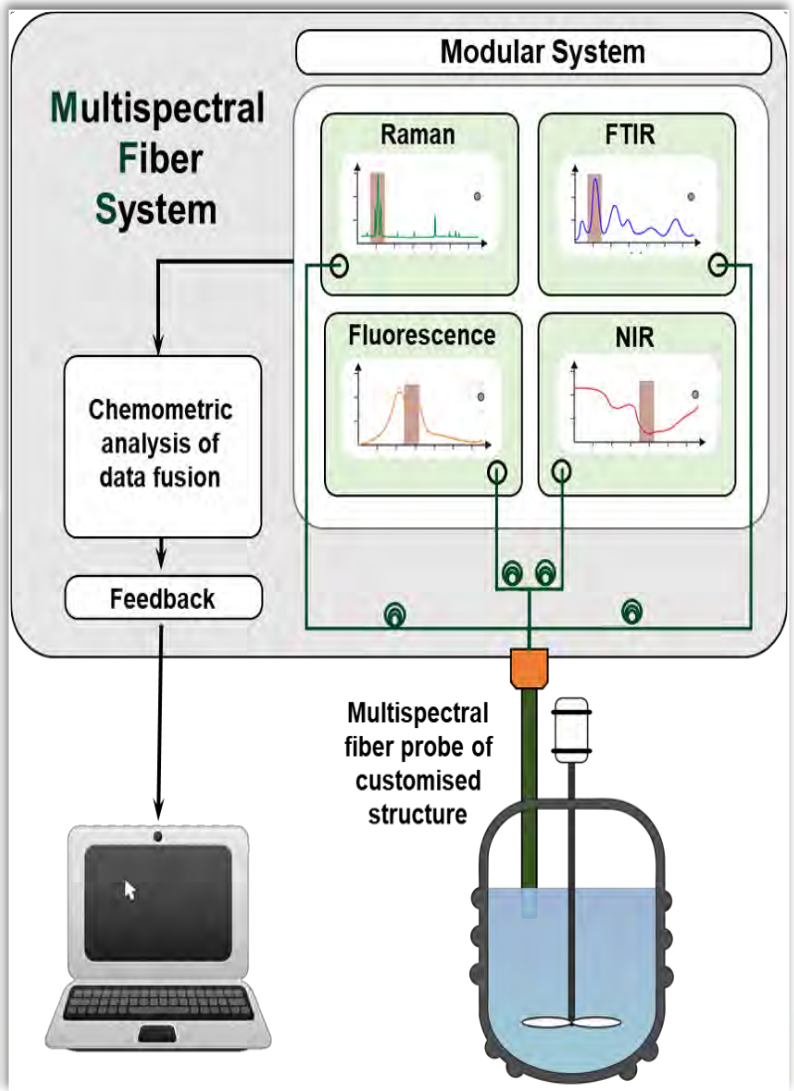



# Choice of optimal System & Method for process-control *in-line*

ArtFiberSystems
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**Multi-Spectral Fiber Systems** allow to test 4 spectroscopy methods in 0.3-16µm range - using robust fiber probes for remote monitoring of chemical reaction in-line. Comparison of spectral data for customer process enables to select the best process-control solution




[LinkedIn post from 9´25 from Andrew Anderson]



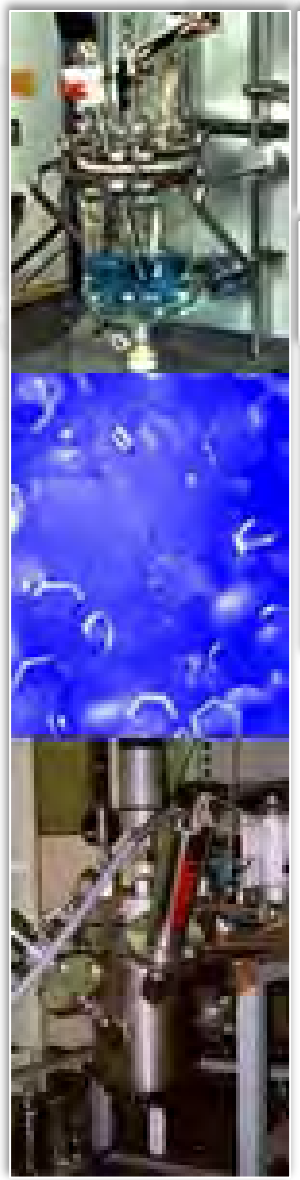





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# 20 years History of Multi-Spectral Fiber System from Martin Warman

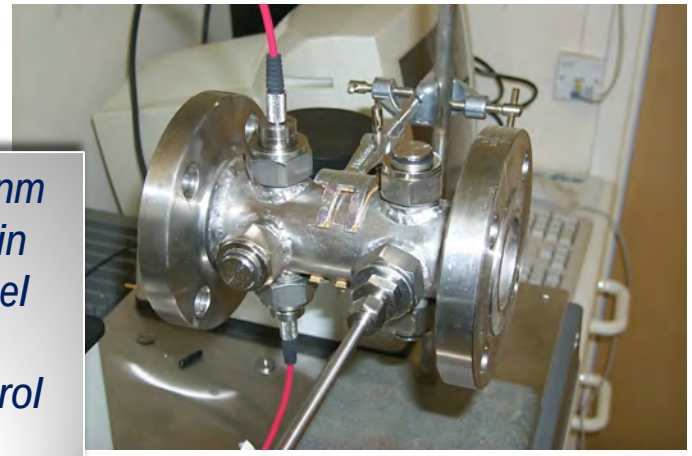
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- **Process Analytical Technology (PAT) - Joint Session**
- **Use of PAT During Pharmaceutical Development and Manufacturing** - Martin Warman, Pfizer

**Real Time Analytics User Forum of Mettler-Toledo**
*London, UK, February 29th- March 3rd, 2004*

…use of 'live process spectroscopy' is not new, and it isn't used because of a technology limitation, in many cases it is simply used for development, so when the process is stable it is not needed - as a simple sensor version is adequate in commercial production



*Flow cell with 254nm LED and detector in C276 stainless steel housing used for transmission control changes with time*

EQUIPNET

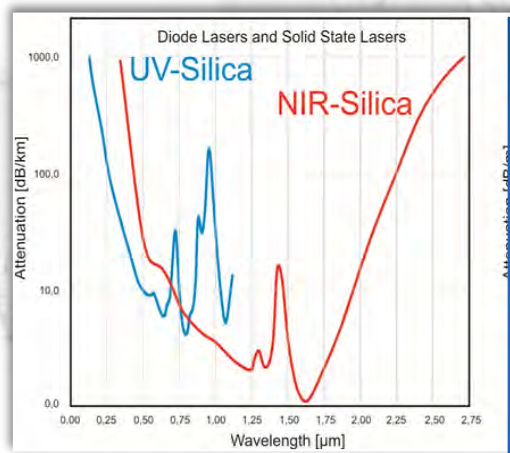


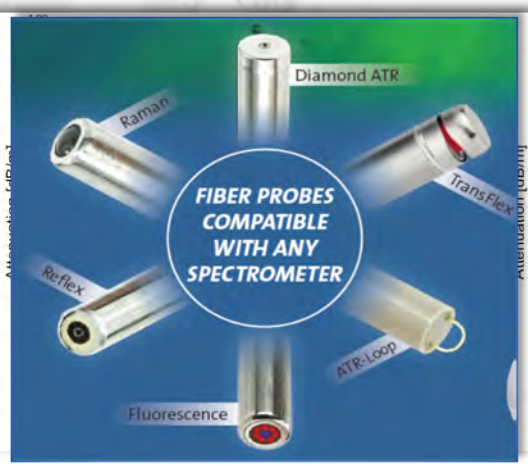


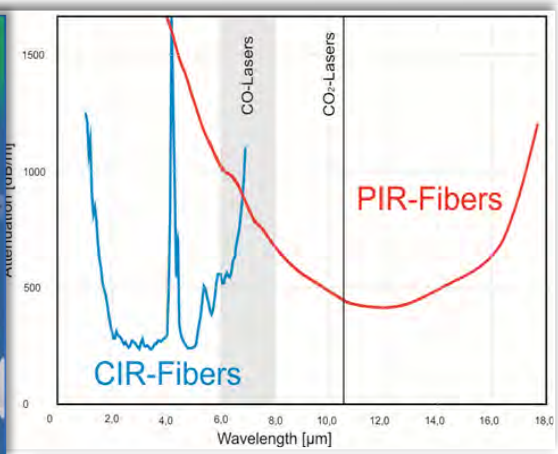

SpectrAlliance Nova Process Spectrometer
Listing #848053 (see equipnet.com)

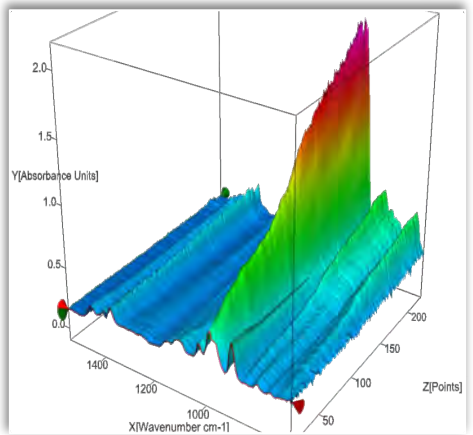
# Muli-Spectral Fiber Systems to select the Best Process Control Method

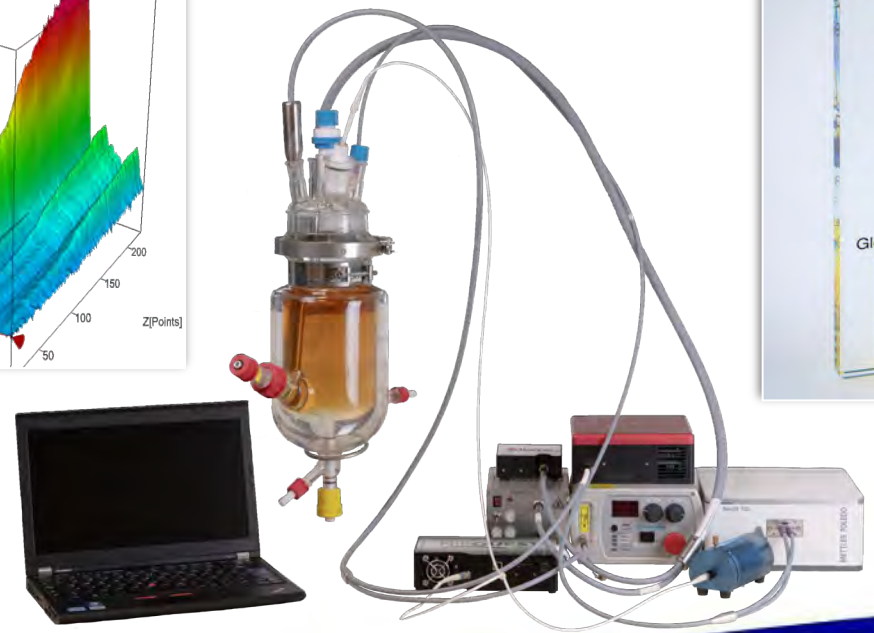
art photonics



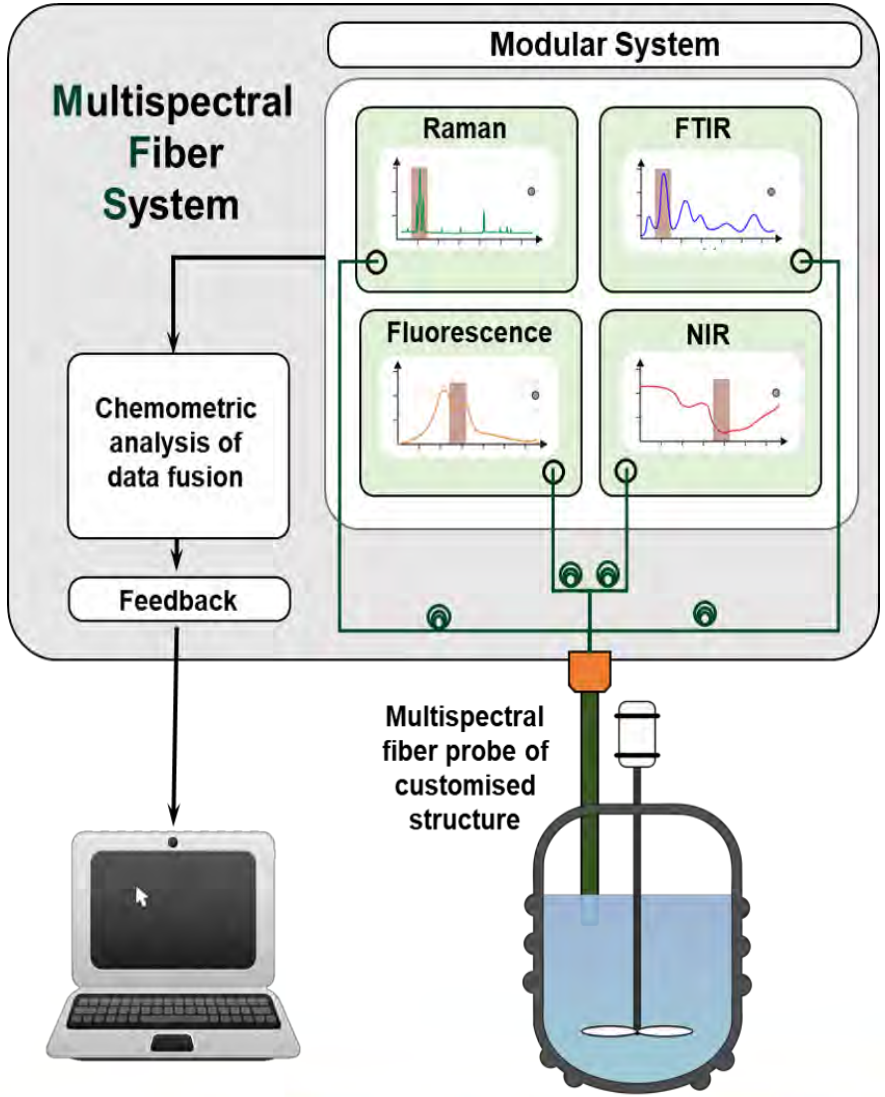



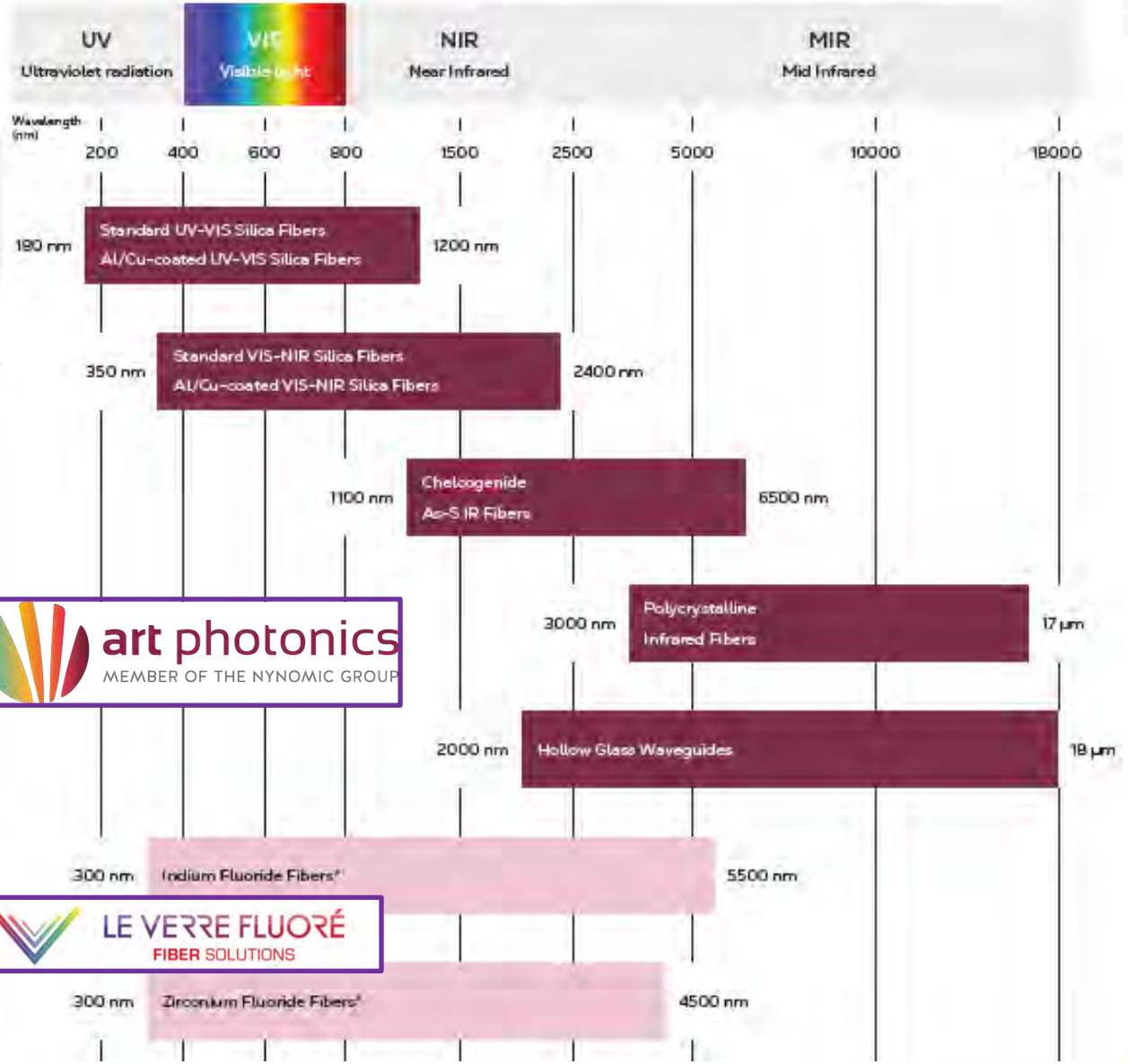





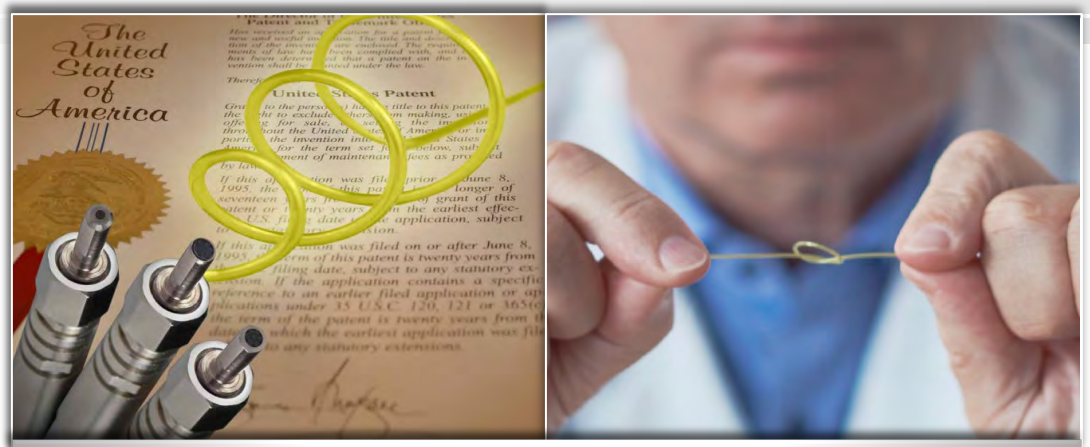
# Transmission for Different Fibers in 0.2-18µm - used for Fiber Probes

art photonics



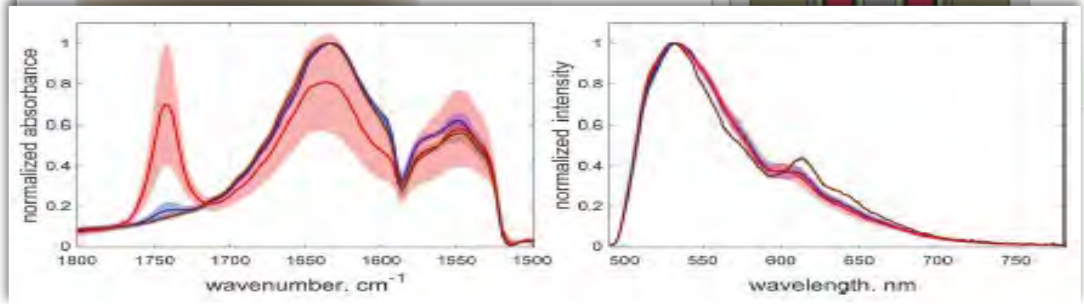

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MEMBER OF THE NYNOMIC GROUP

LE VERRE FLUORÉ
FIBER SOLUTIONS




## Polycrystalline AgHal PIR-Fibers & Combi-Probes from

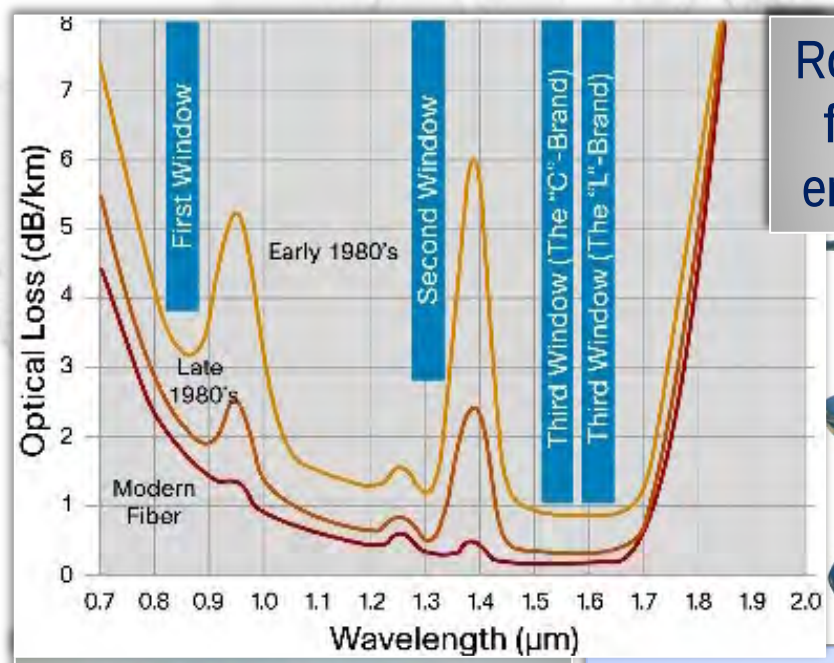


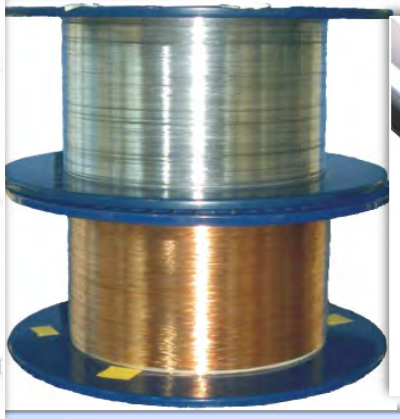

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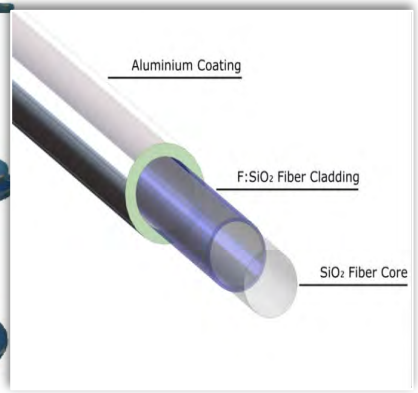
# Spectral Vis-NIR-Cables & for Industrial Process Control

art photonics




Robust 100-300m cables made with Silica fibers of low OH-content (<0.1ppm OH) enable remote industrial process-control








## Alu- & Cu-alloy coated fibers

- High temperature resistance
- Increased durability, high strength, hermetically sealed
- Enables soldering with connectors, bundles, plus to the hermetic inlets
- Free of Polymer jacket fluorescence for Raman and Fluo- Spectroscopy

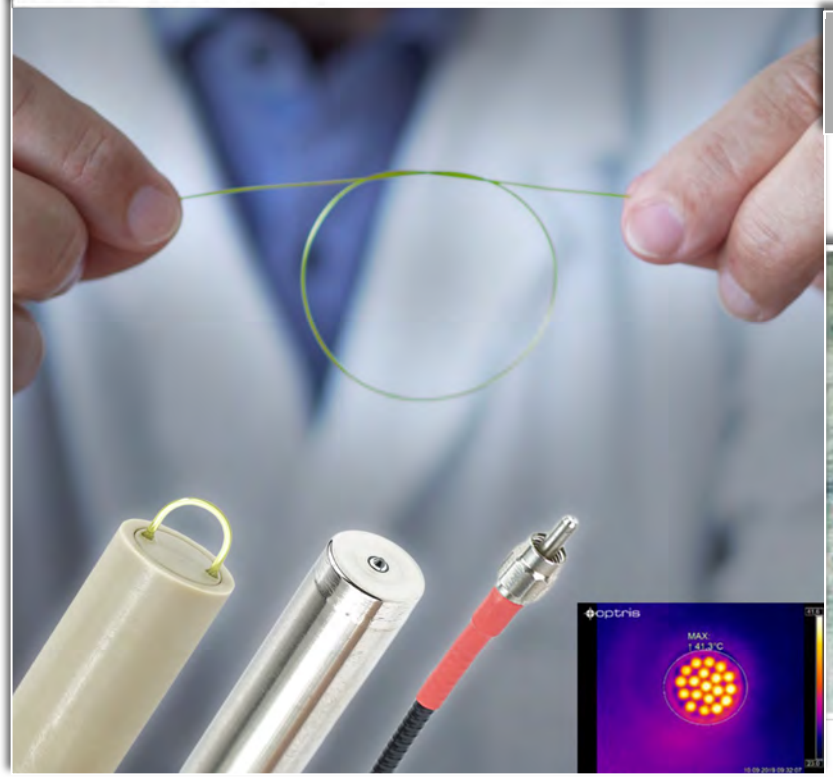


art photonics

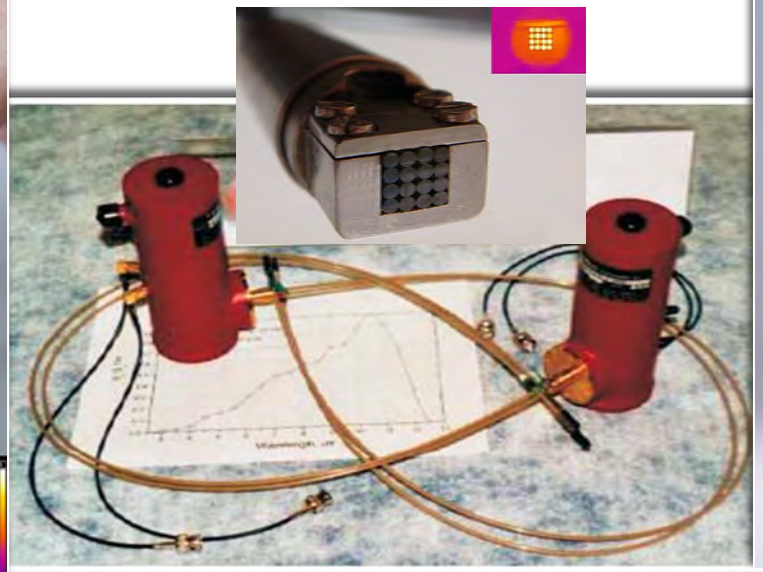
# Global Leaders in Mid-IR Fibre Optics 2020

**EU Business News recently awarded art photonics GmbH with the title of Global Leaders in Mid-IR Fibre Optics 2020 in the German Business Awards. We caught up with this forward-thinking fibre optics company to discover how they came to be worldwide leaders in the field of fibre photonics.**



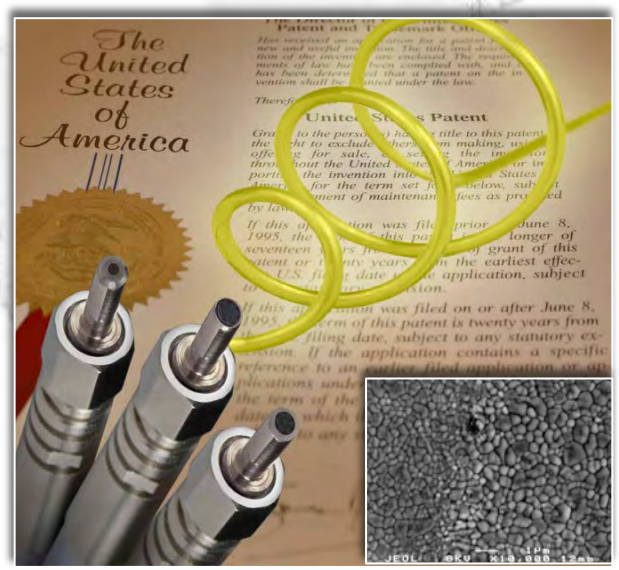

PIR-fiber transmission range 3-16µm is in the perfect match with
Black Body Spectral Radiance at 0-100°C and Finger-Print Region of Molecular Vibrations

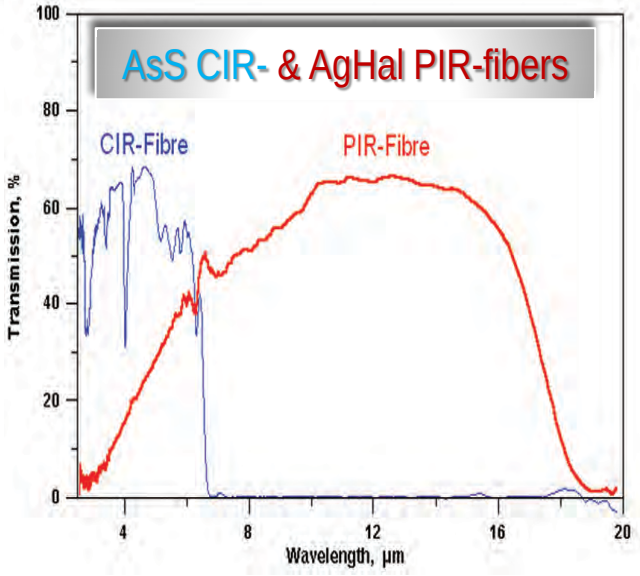


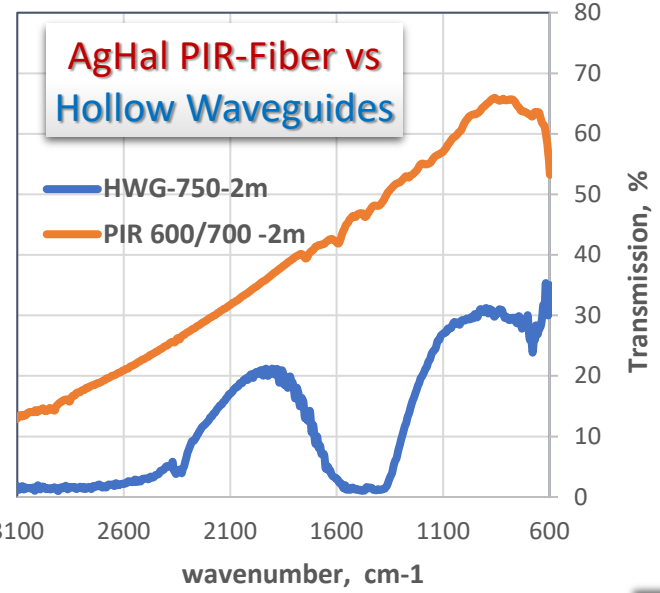


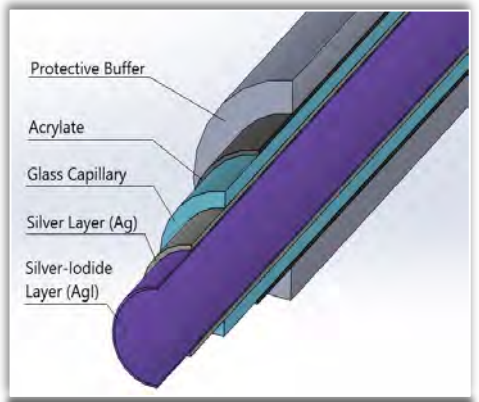
# CIR- & PIR-Fibers vs Hollow Waveguides for Mid IR-Photonics

art photonics



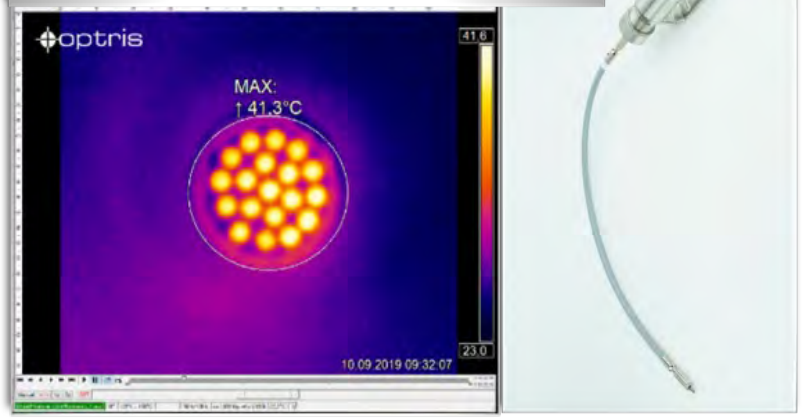



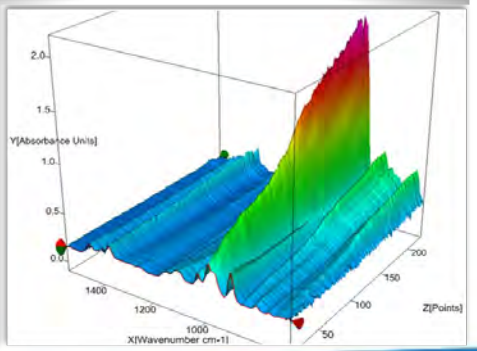

## PIR-Fiber & HWG Cables for QCL, OPO, CO- & CO2-Lasers

## 19xPIR-fiber IR-image bundle





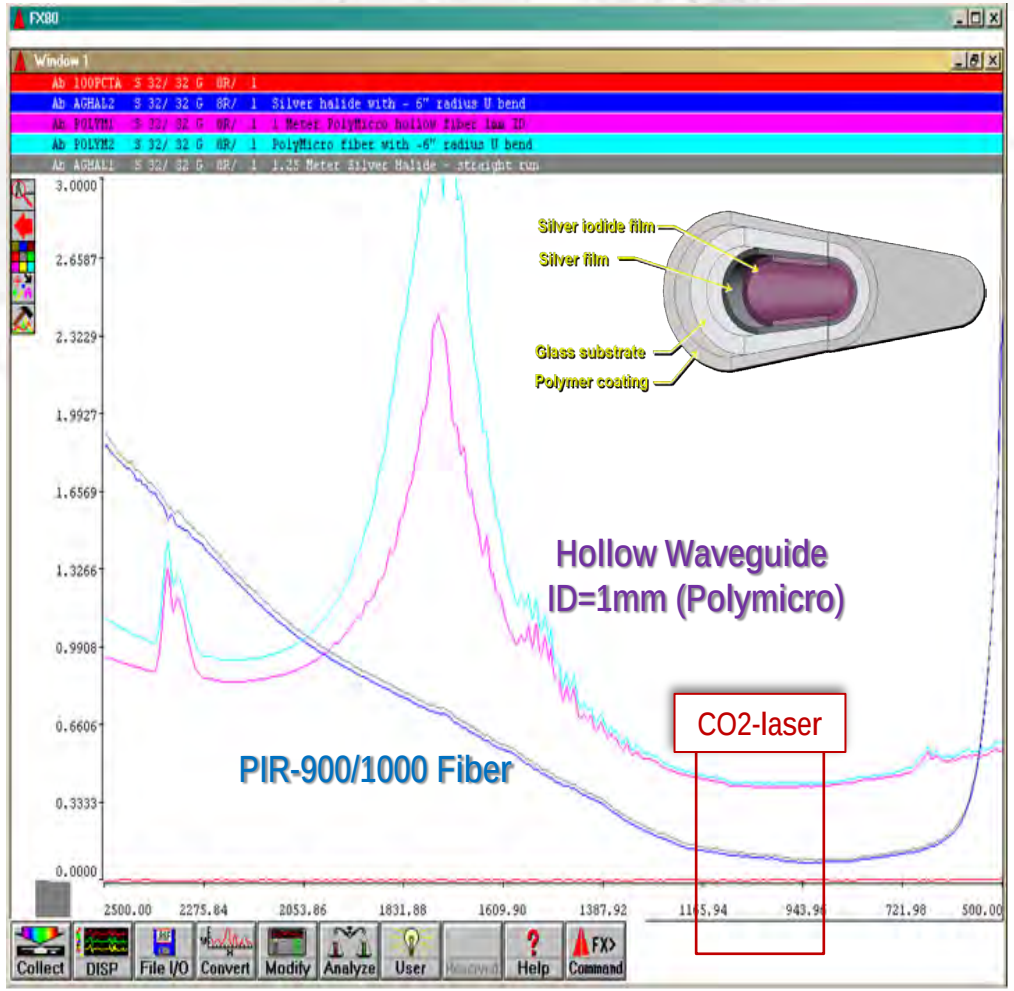

## Fiber Probes for FTIR-Process-Spectroscopy

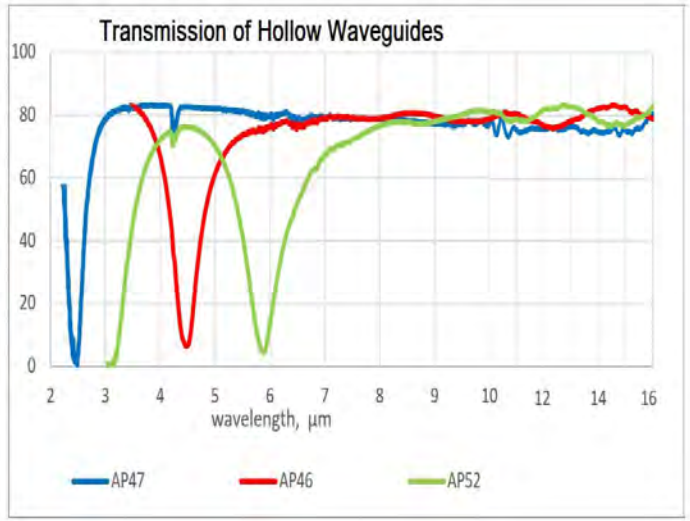




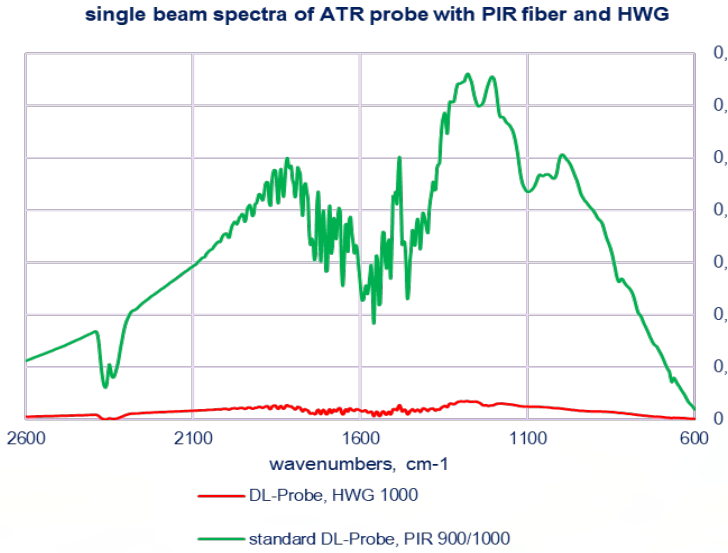
# Selection of PIR-fibers or Hollow Waveguides for FTIR-Spectroscopy

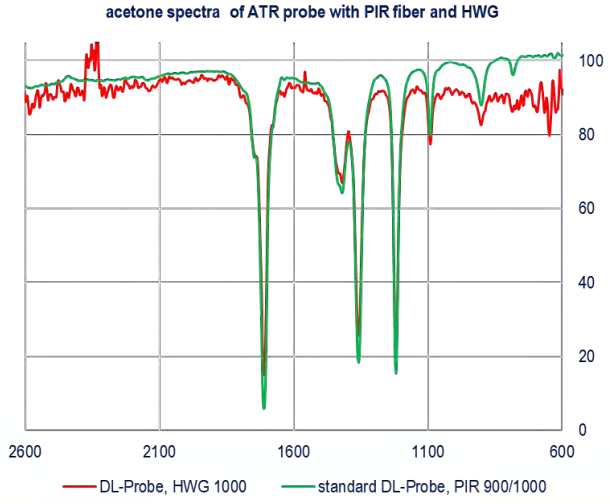
art photonics




**Hollow Waveguides** can be produced with different Transmission spectra, but they possess too low NA>0.1 – which results in very poor coupling efficiency with FTIR-spectrometers, **while high NA>0.3 for PIR-fibers** provides much better coupling efficiency of PIR-fiber based ATR-probes with FTIR-beams- much better vs **Hollow Waveguides**



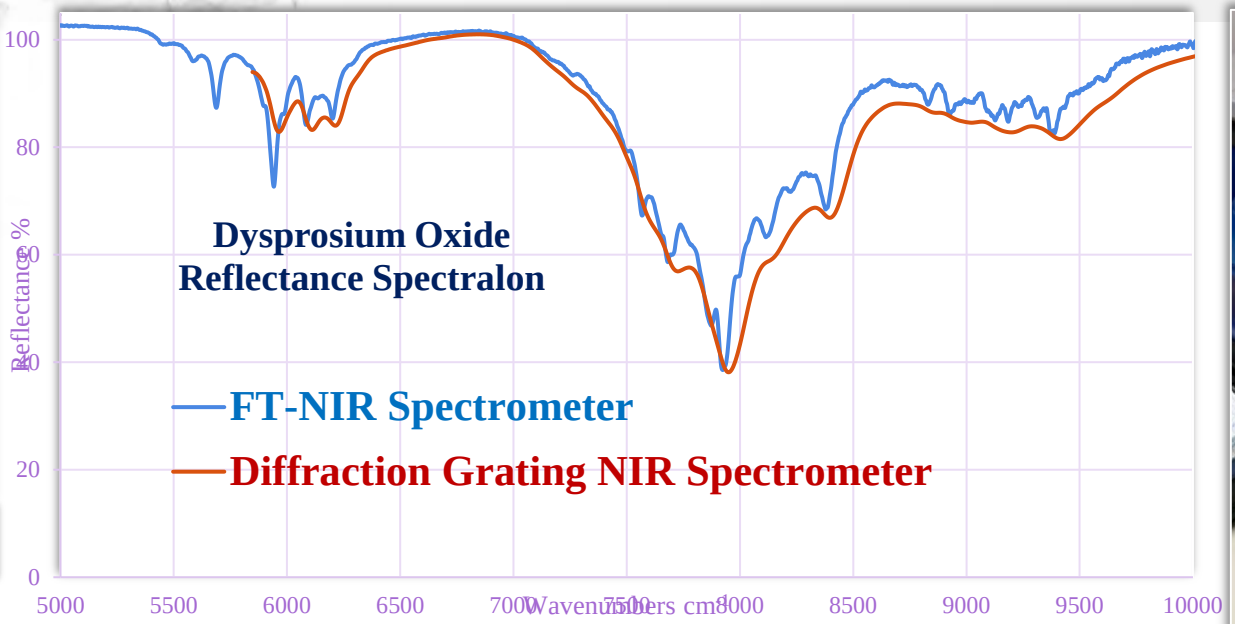

**Very Flexible, non-brittle and non-toxic PIR-fibers are the best in coupling with FTIR-spectrometers!**

# Test of Silica Fiber Reflection Probes with FT-NIR vs DG-spectrometers

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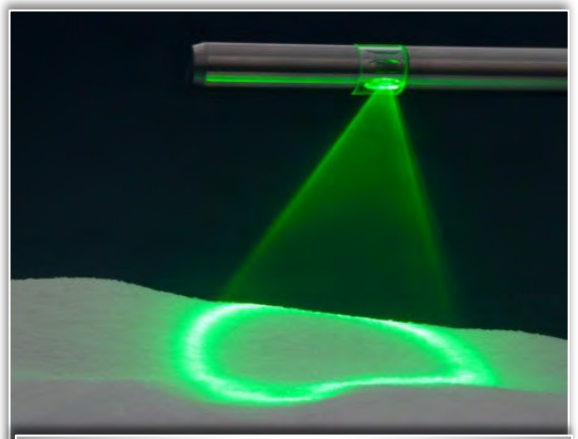
12mm Reflexion Probe







Comparison Test of Reflex Probes with FT-NIR spectrometer Matrix-F (Bruker) and DG-NIR spectrometer NIR-Quest (from Ocean Insight)



Fish-eye DRS-Probe with 2 side windows

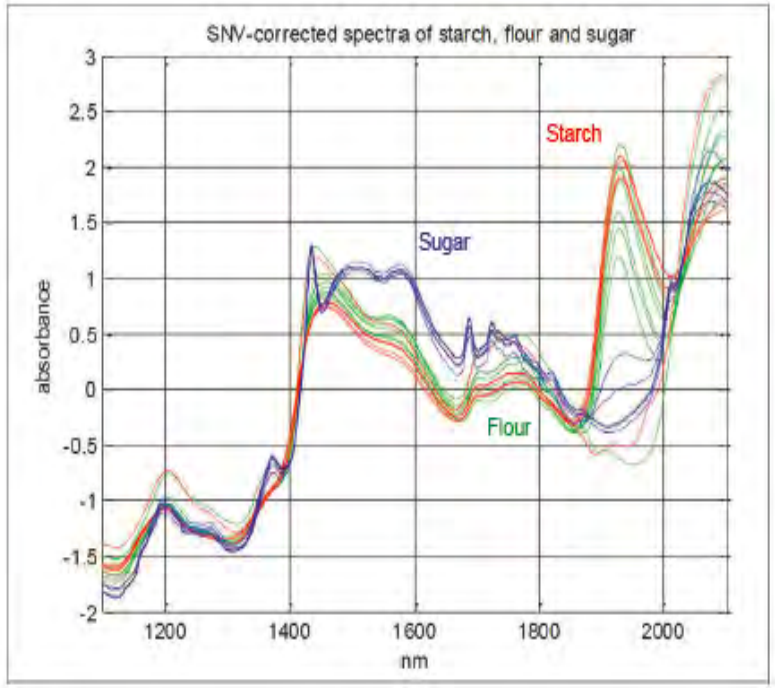


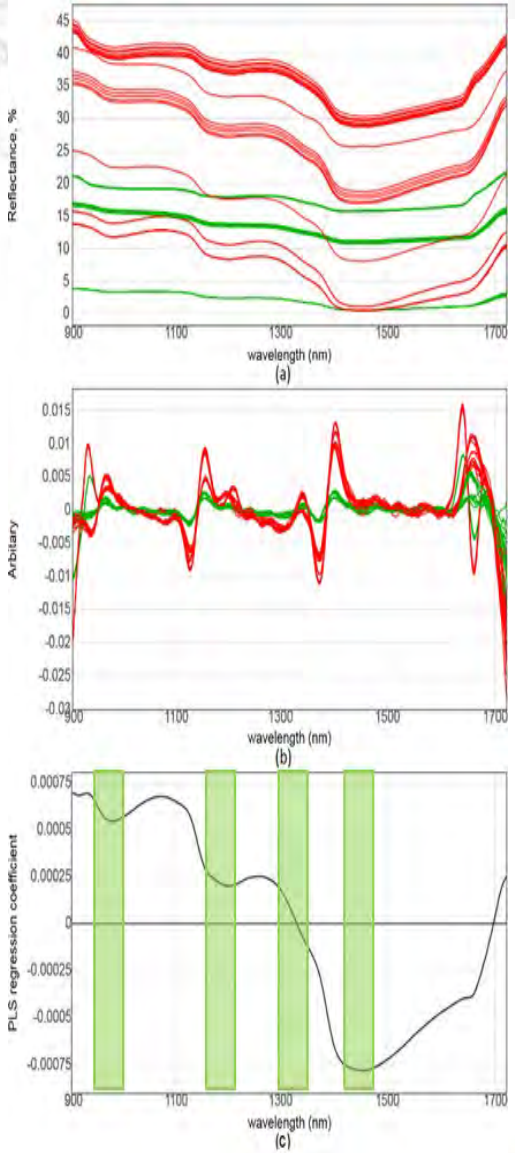
DRS-Probe with 9 side looking Sapphire windows

# IoT Concept of NIR-LED Fiber Sensor to control Moisture Level

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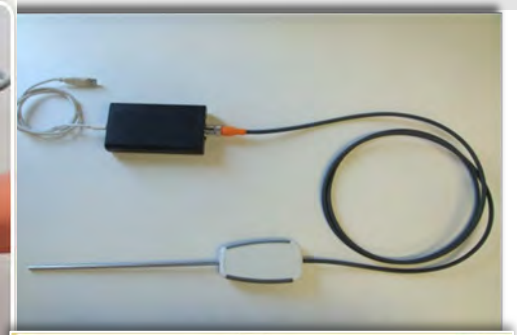


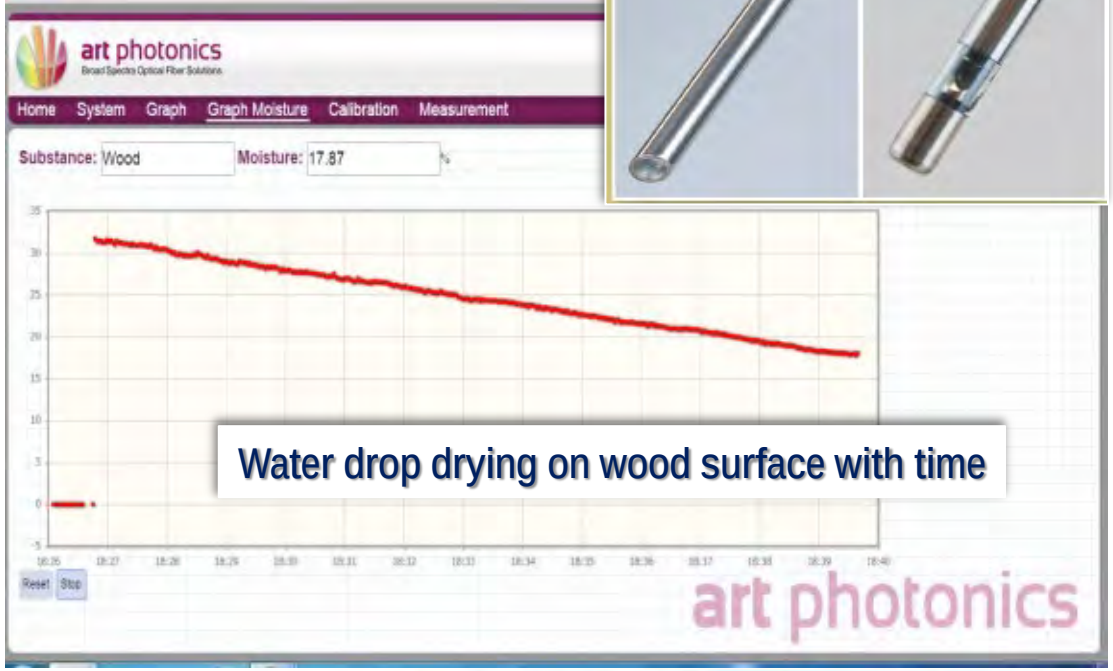











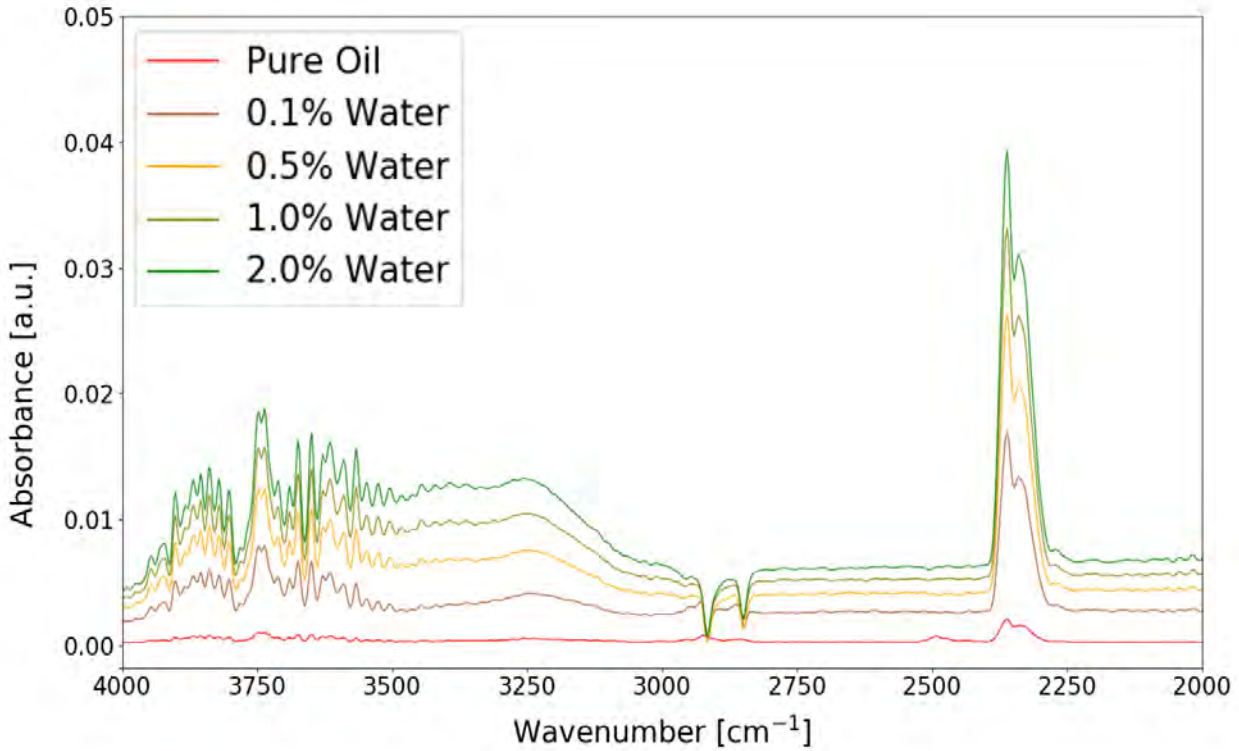


# Application: measurement of water content in petroleum

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**ATR-Probes with ATR-tips from $ZrO_2$ (& Diamond, etc.) sealed in PEEK-shaft and protected by PEEK-tubing can be used in microwave furnaces**



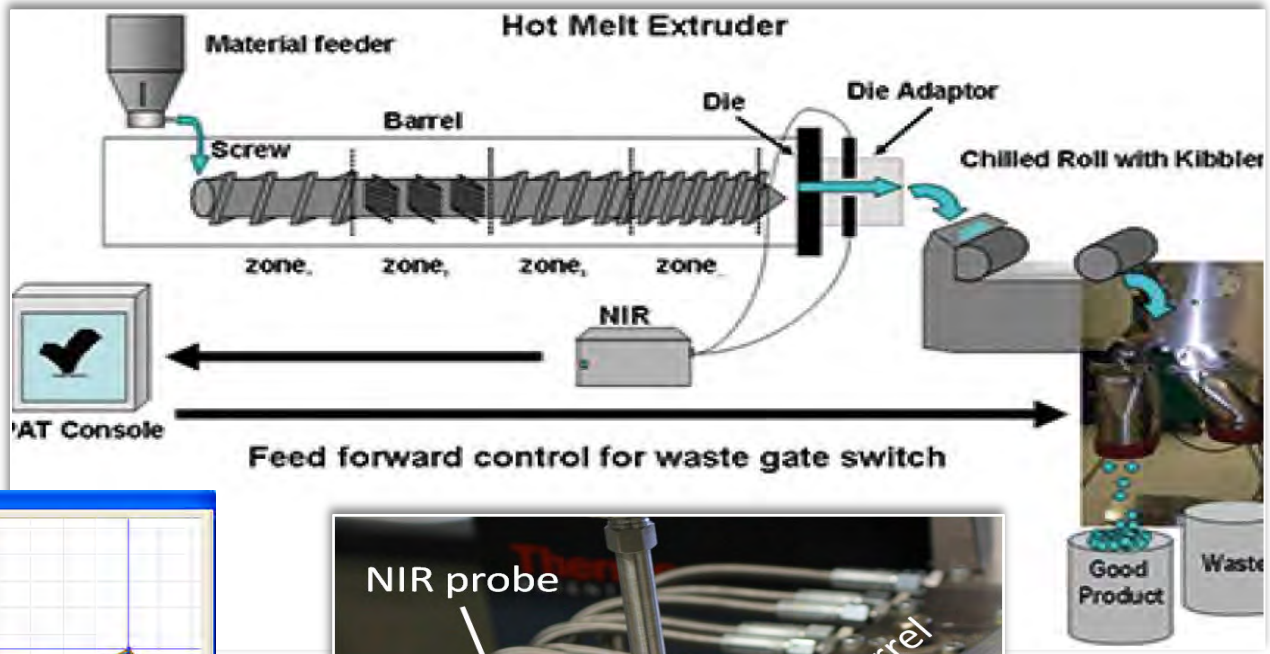


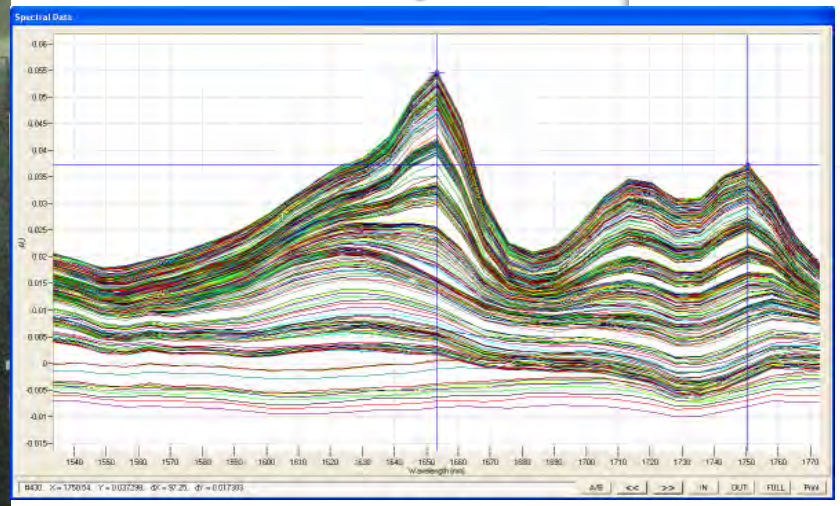
# NIR-Reflex & MIR-ATR-spectroscopy to control polymer extrusion *in-line*

art photonics

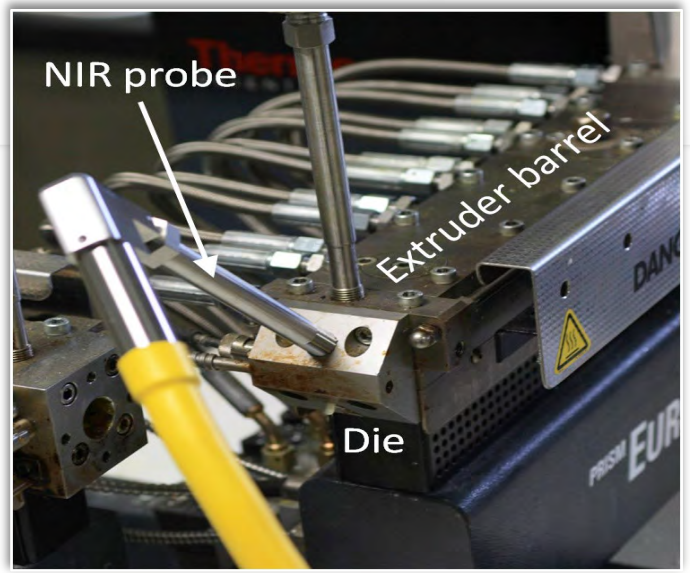


## Control of API-production with PIR-fiber FTIR-spectroscopy in Pharma Industry



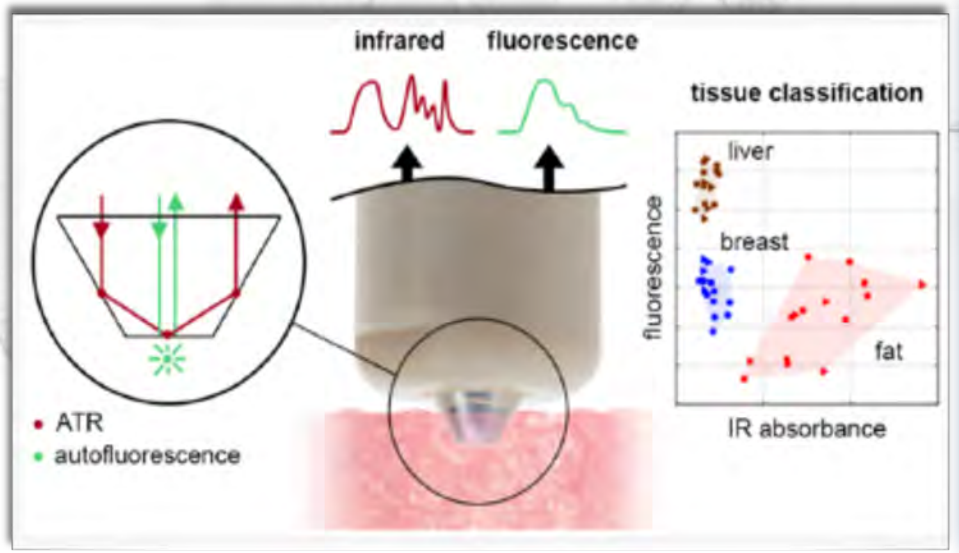



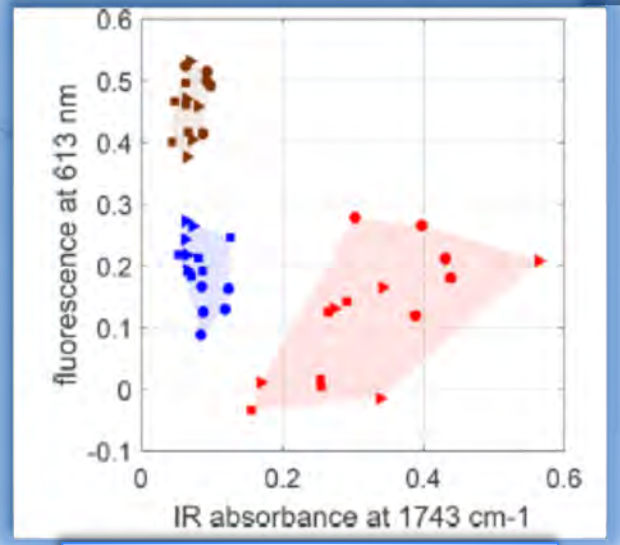
**Spectra of API in hot polymer obtained with Diamond cone ATR probe coupled to iC10 FTIR (MTAC)**

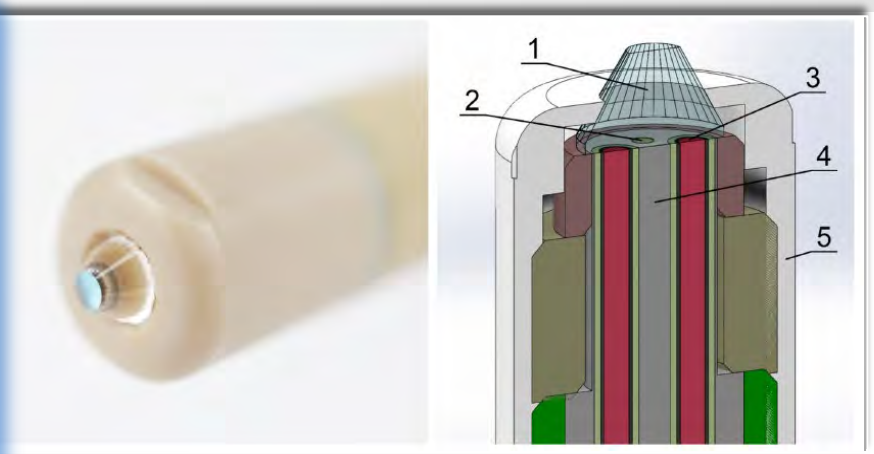


# Sensing with Combi-Fiber Probe to Fuse Mid-IR + Fluorescence Spectra

art photonics



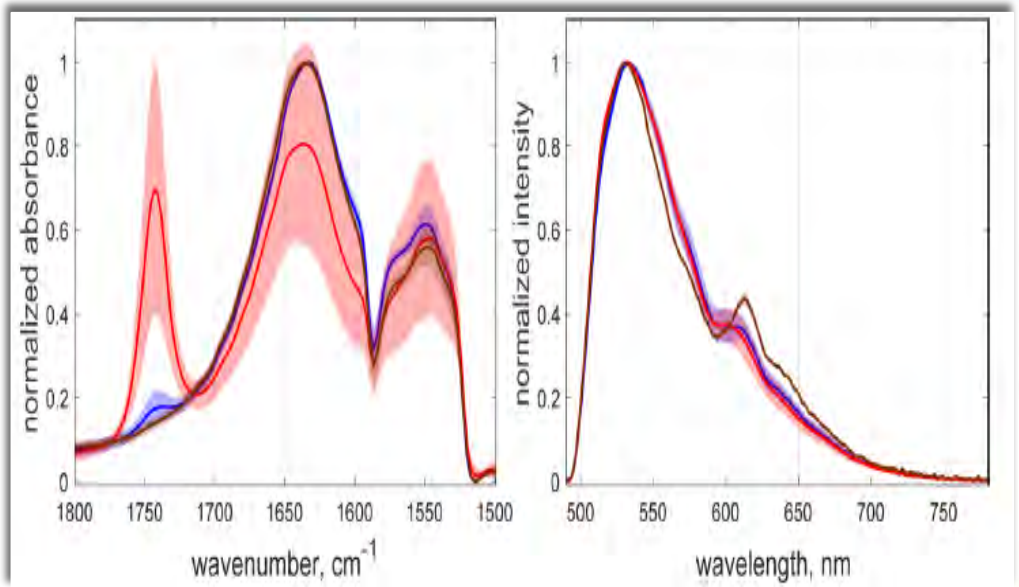



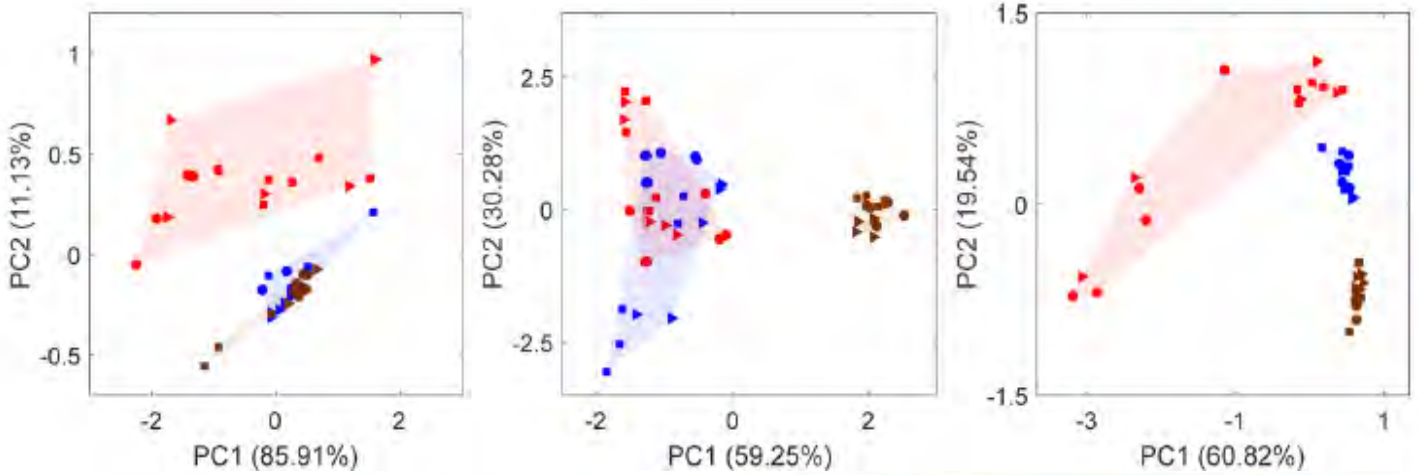




https://doi.org/10.1021/acs.analchem.1c00080

analytical chemistry

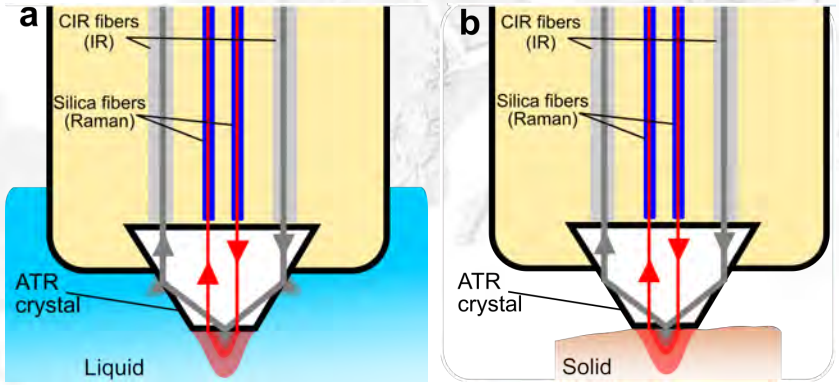
Only 2 wavelengths!



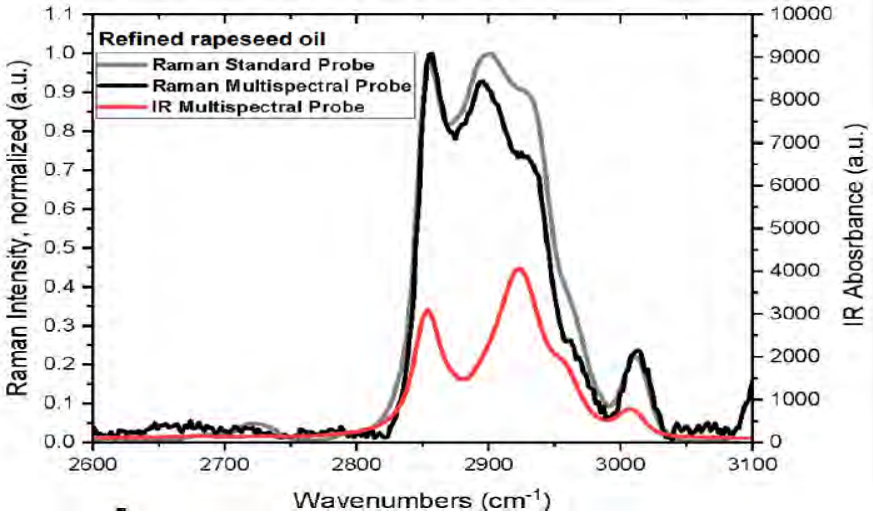



# Combi-Fiber Systems for ATR+Fluo/ Raman spectroscopy

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Combi probe with 2 IR-fibers coupled to FTIR-spectrometer and 2 Silica fibers coupled to Fluo or Raman spectrometers to work with 680nm & 785nm laser




Raman & MIR spectra for refined rapeseed oil



Combi-ATR/ Fluo&Raman Fiber System with Bruker ALPHA, NLIR & Wasatch spectrometers

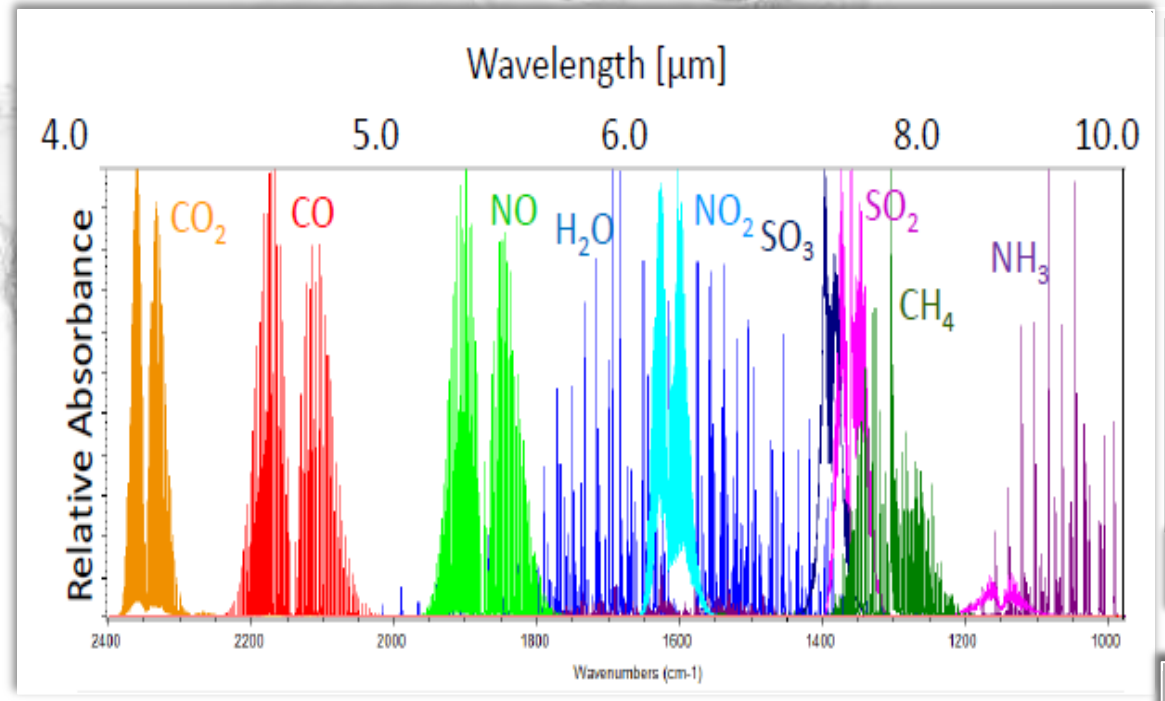
*Nature Sci. Rep. 14, 7430 (2024)* *https://www.nature.com/articles/s41598-024-57539-4*

**Multimodal fiber probe for simultaneous mid-infrared and Raman spectroscopy**


Alexander Novikov,[1,2*] Stanislav Perevoschikov,[1,3] Iskander Usenov,[1,2]
Tatiana Sakharova,[1] Viacheslav Artyushenko,[1] Andrey Bogomolov[4]

# IR-Fiber coupled IC & QC-Lasers for Biomedical Diagnostics

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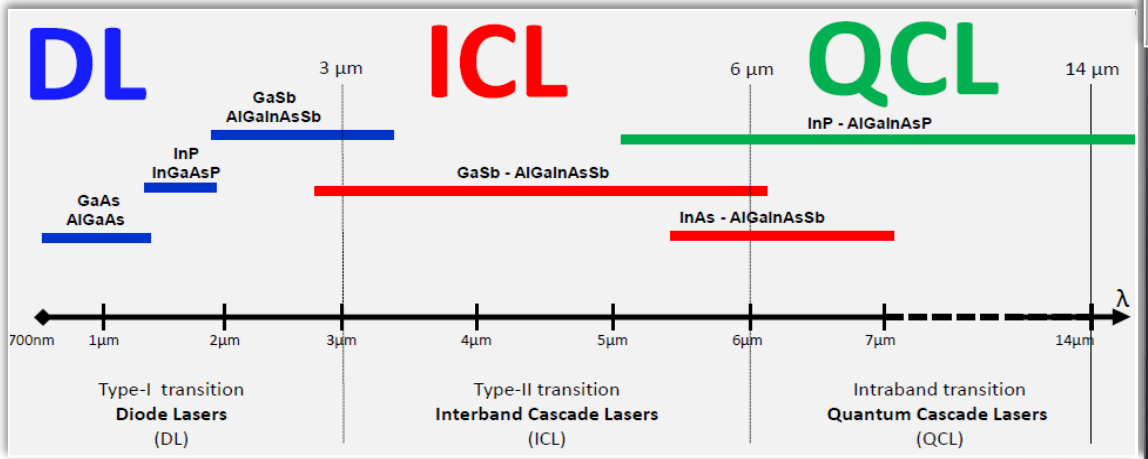


MIRACLE
miracleproject.eu

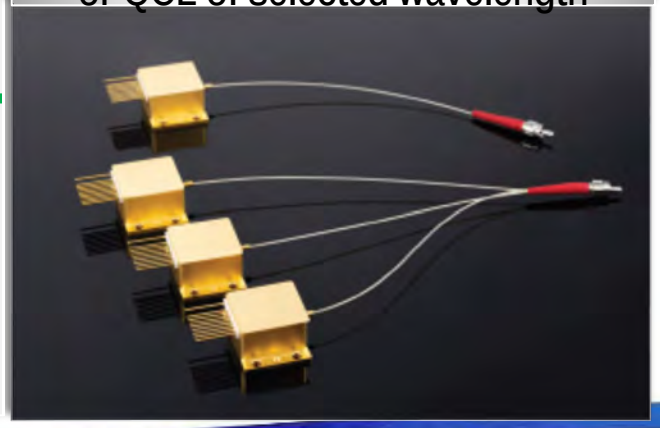
Opto Precision



PIR-fiber bundle combines radiation of 7 QC-Lasers into Arthro-Probe




IR-fiber bundle combiner for any ICL or QCL of selected wavelength



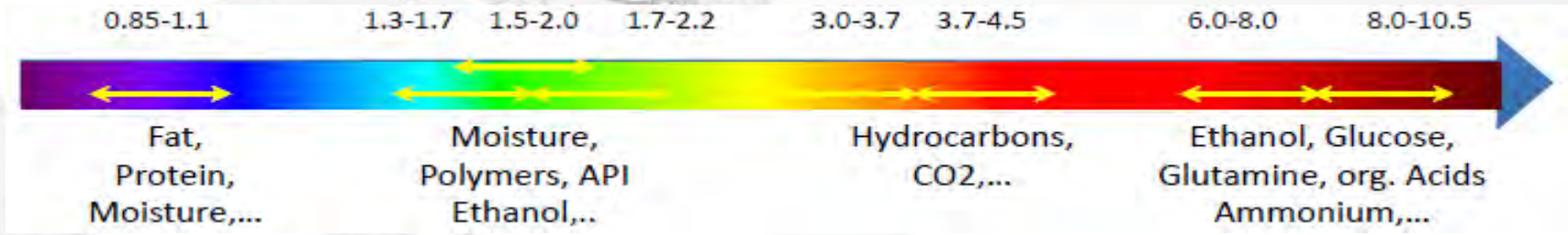
QCL with detachable IR-fiber cable

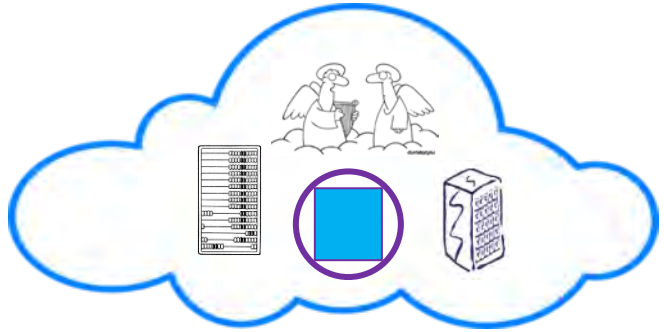



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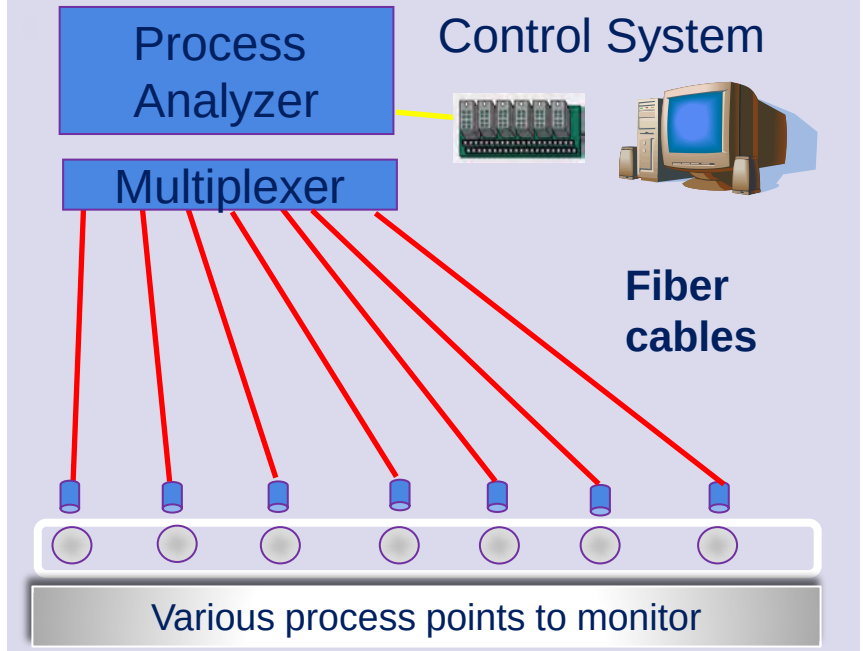
# Spectral IoT Sensors vs Muliplexed Spectral Fiber Analyzer

ArtFiberSystems
FLEXIBLE. LIGHT. SOLUTIONS.



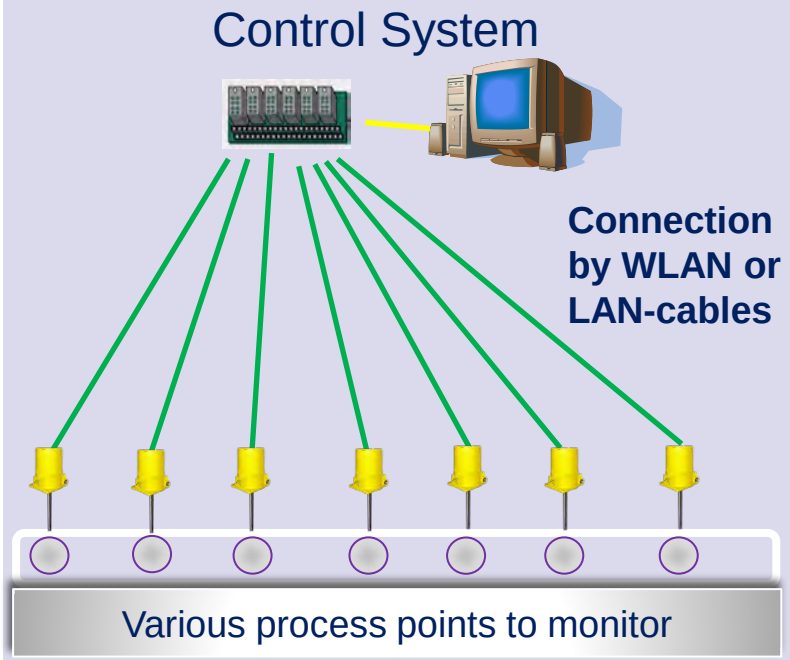



Centralized installation with analyzer/multiplexer and long fibers for measuring points




Advantage - High accuracy due to the higher performance of full spectra process analyzer

Sensors installed in measuring points and send results via wires, Ethernet, Wi-Fi, etc.




Advantage - Lower cost for installation

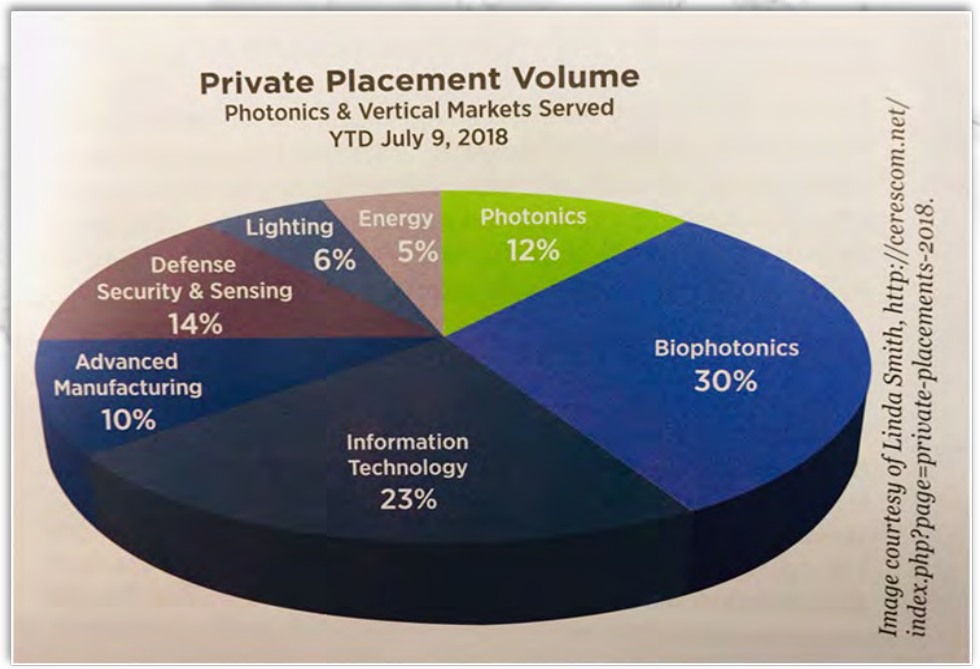
**Innovative Fabry-Perot tunable filters from InfraTec, LED, Filters, IR-LED from NanoPlus, spectral chips from MantiSpectra, etc. enable to develop compact spectral fiber sensors of customized design with data send to iCloud as in IoT - to enable remote process control in customized applications**

The global market for IoT networking solutions will grow from $392.1 billion in 2017 to $1.0 trillion by 2022 with CAGR of 21.6%

www.afs.art

# Market for Photonics Sensors & Detectors ($ MILLIONS)

art photonics



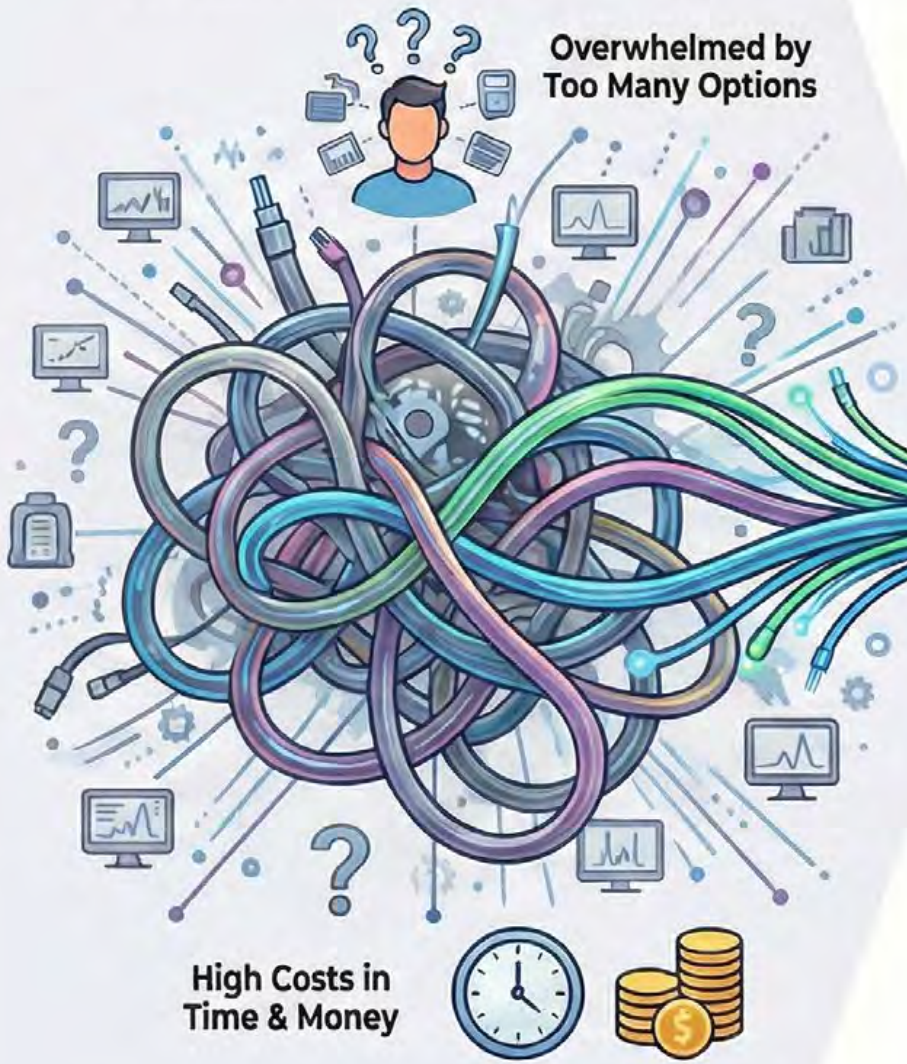

| Application Type | 2015 | 2016 | 2021 | CAGR% 2016-2021 |
|---|---|---|---|---|
| Military | 2,709 | 3,051 | 5,694 | 13.3 |
| Homeland security | 980 | 1,126 | 2,279 | 15.1 |
| Industrial process | 739 | 861 | 1,868 | 16.8 |
| Factory automation | 623 | 730 | 1,635 | 17.5 |
| Civil structure | 645 | 740 | 1,498 | 15.1 |
| Transportation | 566 | 661 | 1,510 | 18 |
| Biomedical | 462 | 540 | 1,183 | 17 |
| Microfluidics | 412 | 483 | 1,084 | 17.5 |
| Bio-environmental | 260 | 309 | 737 | 19 |
| Wind-energy turbines | 226 | 269 | 648 | 19.2 |
| Oil and gas | 254 | 295 | 619 | 16 |
| Others | 174 | 204 | 445 | 16.9 |
| TOTAL | 8,050 | 9,269 | 19,200 | 15.7 |

- **Spectral fiber sensors must be developed for customized applications in industrial process control – to be installed in critical points of reactor for low cost and *in-line* control of reagents composition. Sensors should transfer data by WiFi to iCloud for real time data treatment – to enable process control and its automatization matching the IoT & Industry4.0 concepts**

BCC Research

# Simplify Your Process Monitoring: The Art Fiber Systems (AFS) Pathway

## The Challenge: Complex Process Spectroscopy

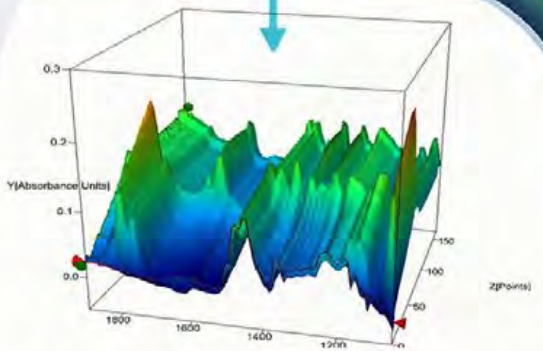



## The AFS Guided Pathway



### 1. Multi-Method Lab Trials

AFS tests your process using all key fiber optic spectroscopy methods.



### 2. Identify Key Spectral Features

We pinpoint the most information-rich data for monitoring your specific reaction.

## 3. Choose Your Custom Solution

Based on the findings, select one of three paths forward.

### Path A: Chemometric Model



Transforming spectral data into real-time concentration changes for process automation.

### Path B: Customized MSF-System



Getting a tailored set of spectrometers to use for R&D in your own lab.

### Path C: Customized IoT Sensors



Developing volume-ready Spectral Fiber Sensers (SFS) for specific process control.